Cutting and sanding shells

Being cautious of the dust created, you can cut and sand shells with nearly any type of high speed cutters. These can be carbide dental burrs in a high speed hand piece or a cutoff wheel in a rotary carving tool. Bear in mind that the force on the shell should be minimal, letting the cutter perform all of the work.

When working with shell, the higher the speed and lighter the touch work the best

Colouring shells

Colouring can be easily done with acrylic spray paint, egg dyes, alcohol markers and airbrush colours

Taking advantage of the shell porosity, 'magic' marker colours can be wicked into the shell

The techniques for colouring shells fall into two different categories depending on the shell type. For all egg shells and non-seamed sea shells, you can use nearly any immersion dye or applied paint method. The texture of the shell surface will provide a great substrate for acrylic paints to bond to. If you intend to do any piercing or shell alterations, it is usually best to do those before dyeing or painting. That way the open edges will be covered with colour and won't present the tell-tale white, showing your alteration edging. You can also take advantage of the porosity of egg shells with any open edges. Alcohol-based dyes will wick into the shell from these surfaces creating a unique colouring effect. For shells such as the sputnik sea urchin shells, caution must be used when colouring. Any immersion process will likely cause the shell to separate along one of the seams. Any coloration should be done with an additive spray process with minimal shell wetting.

Bonding wood to shell

In a perfect world, you'd never bond shell to wood based on their dramatically different characteristics. Wood will continue to take up and give off moisture forever depending on the relative humidity. That moisture change will cause dimensional changes. Shell will not be susceptible to this like wood. A brittle non-moving material bonded to a dimensionally changing material is a recipe for joint failure. Of course this is tempered with the species, grain orientation, range of humidity swings and the method and amount of interface. You may wish to review *Woodturning* 252 and *Woodturning* 269 for in-depth explanations of where, why and how to bond and adhesives. Whenever possible, I avoid shell to wood bonding. When I must bond the two together, such as the chicken egg shell to the finials or stand, I use epoxy. I take pains to minimise the glue area and effect a glue trap in the wood portion. These two actions will reduce the likelihood of any failure of the joint because of the expansion or brittleness. Another advantage is that most of the wood interface is end grain, nearly always being a spindle turning. While end grain is usually not the selection for an adhesive join, here it works to our advantage.

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